



Salesforce Platform Architecture

Understanding Development

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Ground Rules

Let's agree to some ground rules before we proceed

- Training will be delivered considering you do not know anything about Salesforce. So, even though you have prior experience keep your mind blank and open to learn more effectively.
- This is online live Training Session. Focus completely on session, even though recordings will be made available.
- Please make sure there is No background noise or disturbances.
- Questions will be addressed at the end of the session and will also setup a recurring meetings to discuss that week's question.
- **NO QUESTION IS SILLY.** So, you can ask same question again n again till you understand completely.
- **Phones on Silent.**
- **Complete all the assignments on time so you can understand next session.**
- **Do not forward invites unless you confirm with Trainer first.**



salesforce

Hola Developer!!

Ready to embark on new journey in Salesforce.
Let's go.....





Overview of Salesforce Developer Role

Salesforce developers are responsible for **designing, developing,** and **maintaining** custom applications and solutions on the Salesforce platform.

They work with the Salesforce platform to **automate** business **processes**, create **custom user interfaces**, and **integrate with external systems**.

Salesforce developers need to –

- Understand the requirements and ask questions
- Estimate the effort required for completion
- implement and test the functionality using config or custom approach
- Perform Impact analysis to asses the roll out of new changes
- Creating technical guides and artifacts
- Deploying changes between environments
- Evolve and learn about new Salesforce release features





Core Skills Needed for Developer



General Skills:

- Object-oriented programming (OOP)
- Web Services (SOAP & REST)
- HTML, Javascript, JQuery
- DevOps – Copado, Git

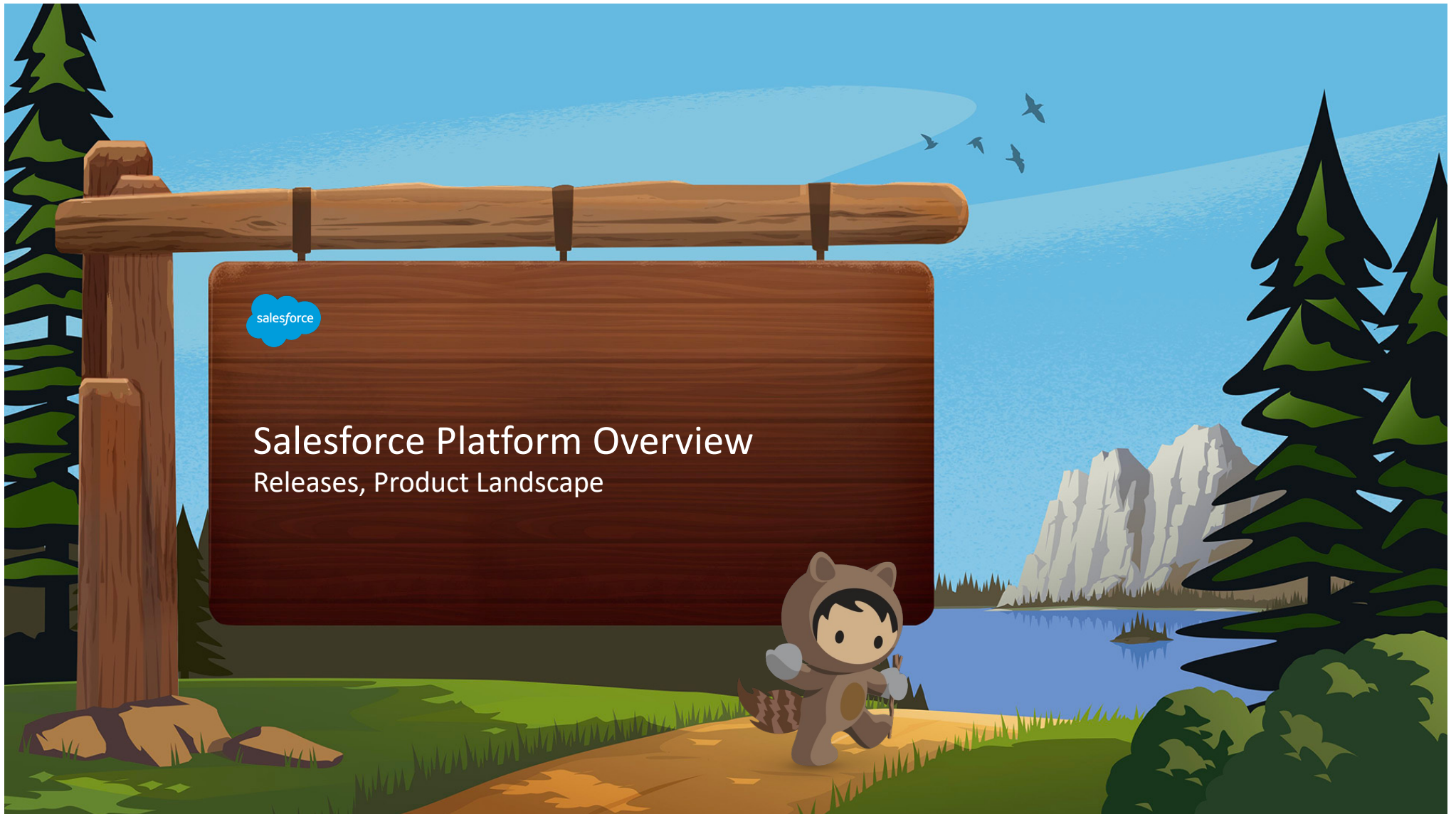
Salesforce Skills:

- Admin basics – Data Modelling, Customizing Lightning Platform
- Apex, Visualforce, SOQL, SOSL
- Lightning – Aura and LWC

Business Skills:

- Communication Skills
- Agile Fundamentals and tools like JIRA, Rally







Why Salesforce?

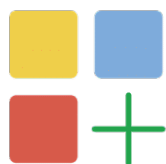
Salesforce offers the world's leading cloud-based CRM software, offering a range of innovative and far-reaching CRM solutions that can be effectively customized to the requirements of businesses of all sizes, from global enterprises to start-up micro-enterprises





Key Terminologies of Salesforce Architecture

Before we dig deeper, let's understand few things.



App



Instance



Superpod



Organization



Sandbox

App : Logical grouping of commonly used features. Use App Launcher to launch the apps provided OOTB and installed from AppExchange.

Instance : Your space in the Salesforce ecosystem. Basically, your server instance with your personalized configuration.

Superpod: Arrangement of Frameworks, Servers and stack balancers, supporting instances.

Organization: single consumer of the Salesforce application.

Sandbox: This is the copy of your production instance, so you do not have to work in Live environment.



Frequently Used Terms in Salesforce

Edition	bundles of Salesforce products and services, each geared toward a different set of business varying by feature, functionality, and pricing.
Organization	A deployment of Salesforce that has a defined set of licensed users. Your organization includes all your data and applications.
Release	Salesforce releases new products and features three times per year, Winter, Spring, and Summer—along with the calendar year. Example: Winter '15. For every Salesforce release, the Salesforce release notes include new features and products that are generally available or in beta release. They also describe all changes to existing features and products.
AppExchange	marketplace featuring hundreds of cloud applications created by Salesforce customers, developers, and partners. Many of the applications are free and are pre-integrated with Salesforce, enabling to easily and efficiently add functionality.
Lightning Platform	platform as a service (PaaS) enabling Developers to create and deliver any kind of business application entirely on demand and without software. Provides easy to use point-and-click customization tools to help create solutions for unique business requirements, without any programming experience.
App	collection of components such as tabs, reports, dashboards, and Visualforce pages that address specific business needs. Salesforce provides OOTB apps which can be customized to fit your needs.
Object	A definition of a specific type of information that can be stored in Salesforce. Objects are representation of Tables in Database and have fields storing the constraints, type of data they can store.
Record	A collection of fields that store information about a specific item of a specific type. A record is an object, such as a contact, an account, or an opportunity. For example, you can have a contact record to store information about Joe Smith, and a case record store information about his training inquiry.

How does Salesforce Release Features and Enhancements?

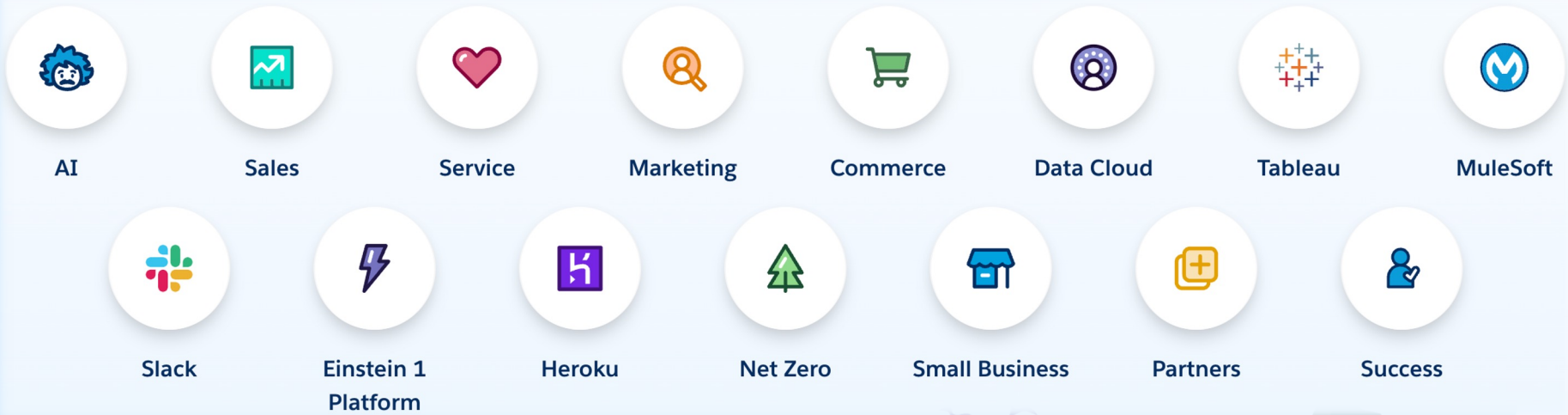
Salesforce releases regular releases three times a year (winter, spring, summer) that include many new features and enhancements.



[Salesforce Release Blog](#)

Salesforce Product Landscape

Helps build Business Specific Applications which are more scalable and reliable.

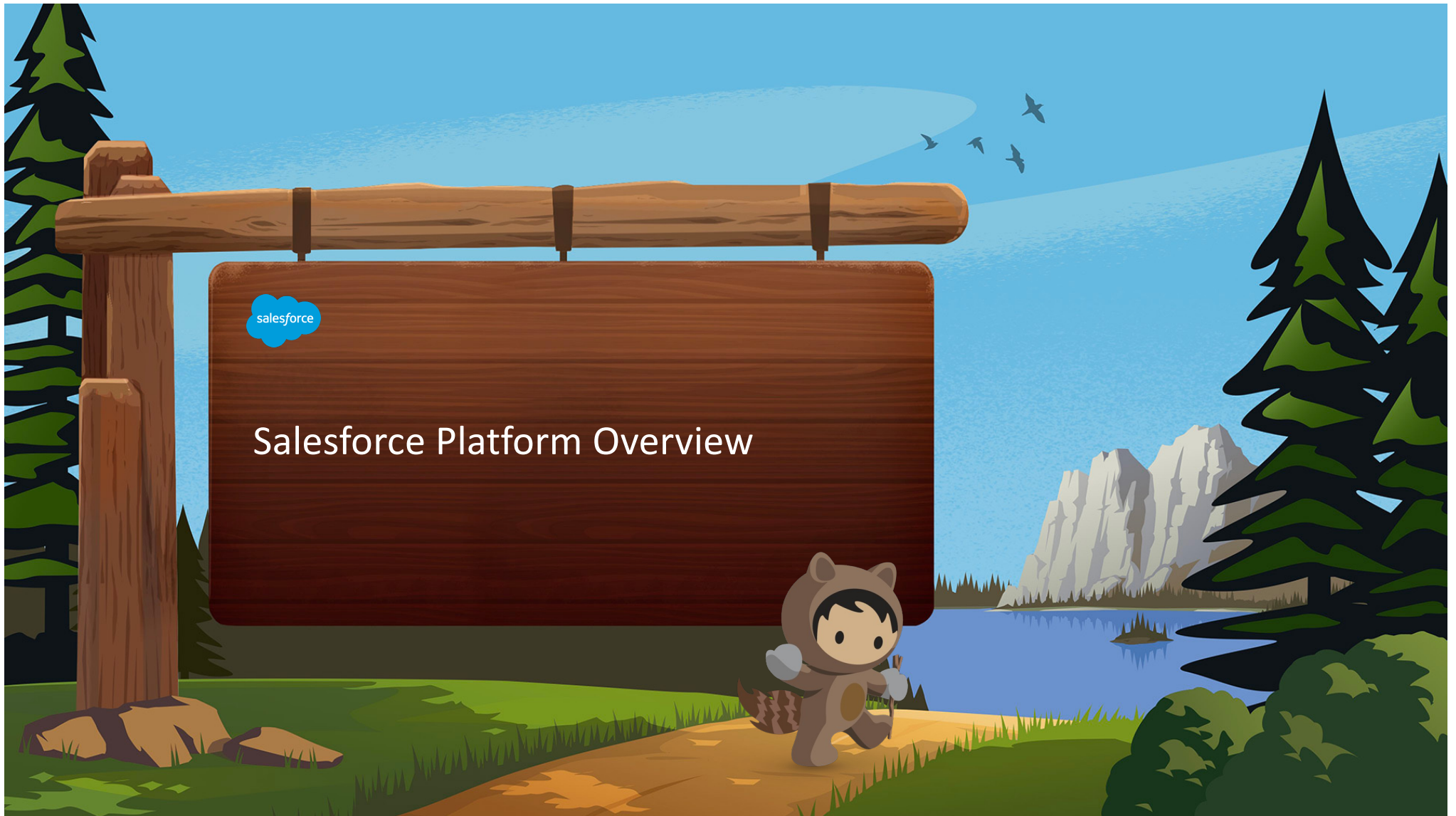


Salesforce Industry Landscape

Helps building Industry-Specific Applications for more specialized use cases. Salesforce also offers trial orgs and demo orgs for these to explore further.

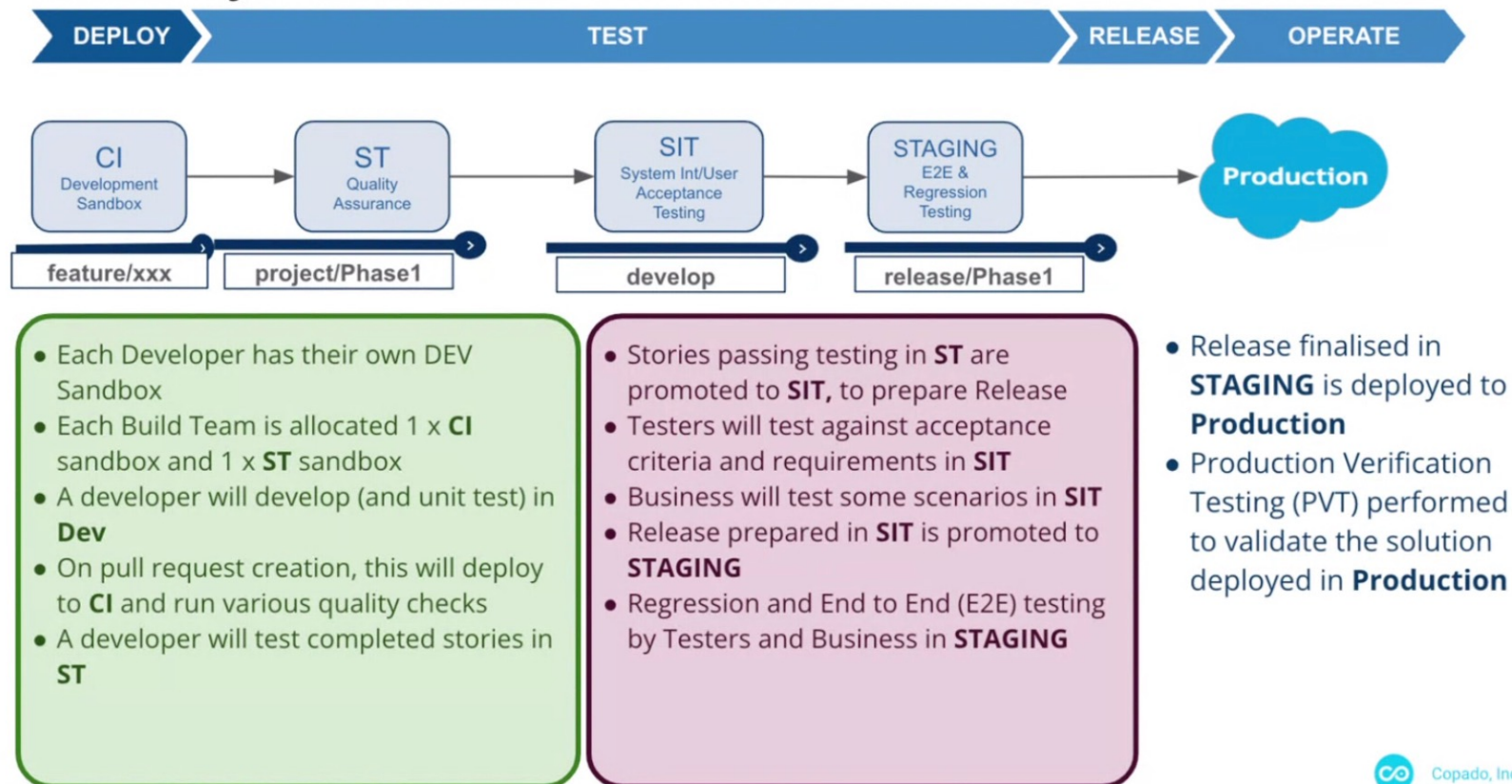
Government	Media	Consumer Goods	Transportation & Technology	Financial Services	Automotive	Energy	Tech
							
Communications	Retail	Manufacturing	Healthcare	Philanthropy	Non-Profit	Education	Sustainability
							
		Public Sector	Grants Management	Brand Management			
							





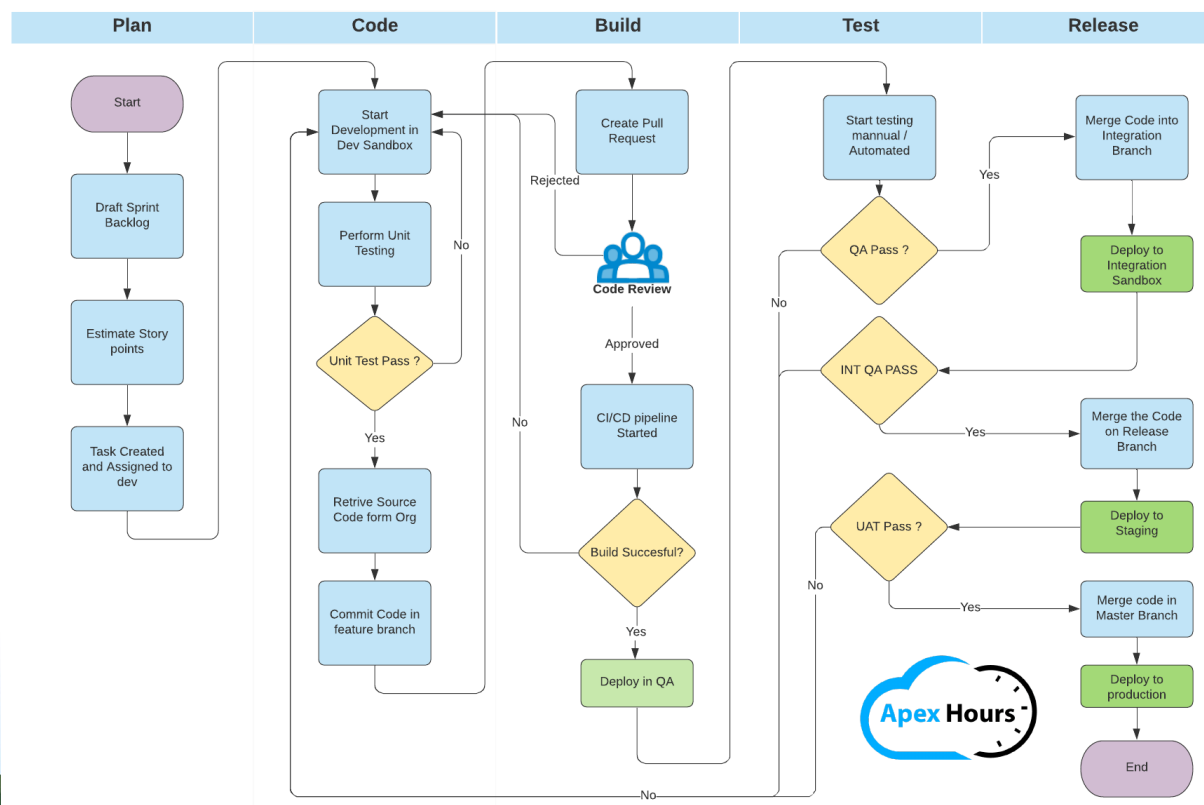
How Enterprise Orgs Handle Projects?

Projects need multiple skills and multiple teams work on them parallelly.



Project Development and Deployment Process

Pushing changes or updates from one deployment environment to another. You will always have your live environment, which is called the live environment or production environment.





Environments (Orgs) in Salesforce

Environments helps securely building applications without overstepping on the functionality built and being used.

Production Environment:

- Salesforce.com environments that have active paying users accessing business critical data are known as production environments.
- It stores the live data that is actively used to run the business and this where end users access Application from.
- **Used by** – End Users, Business Users.
- **Note:** Not all development can be done directly in Production environment,

Development Environment:

- Environment used for development, testing and UAT sign off. Helps prevent mishaps in Production or Live environment.
- Stores all the metadata and some data from Production based on type of Environment.
- **Types of Development Environments** – DE Edition and Sandboxes
- Sandboxes are allocated with Production environment purchase only.

**Instance of Force.com platform
allocated to specific Tenant is
called – An Org or Environment**

**Used for Application development
and storing data and metadata.**

**Uniquely identified based on the
“Org Id” assigned**



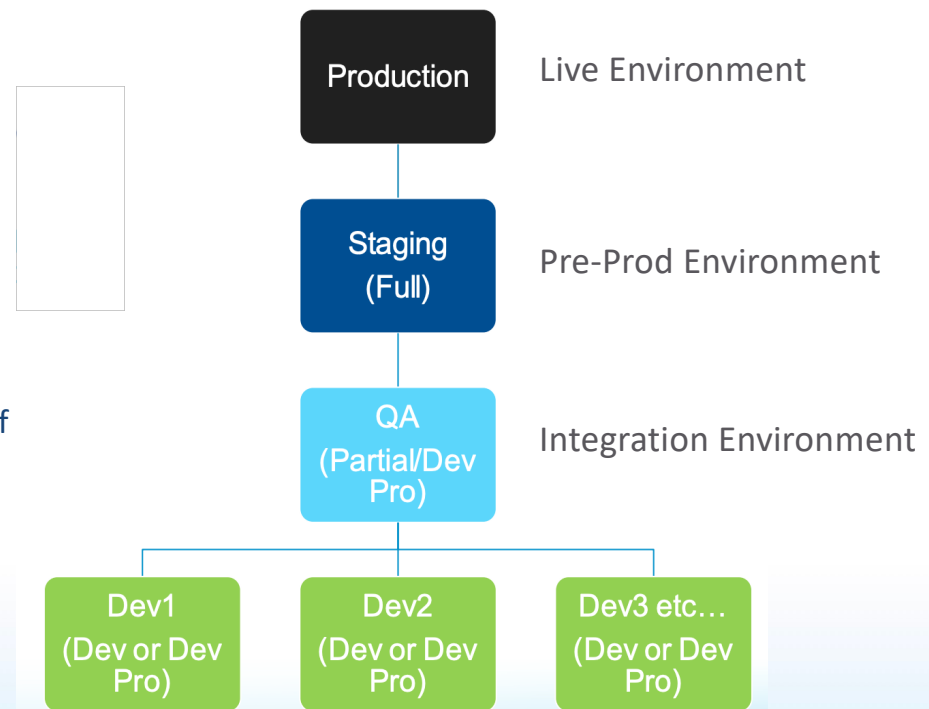
Why we need Orgs in Salesforce?

Editions define the features accessible within the Org purchased.

Production Env – login.salesforce.com

Sandbox Env – test.salesforce.com

- Salesforce Environment refers to an organization that is used by a customer for a specific purpose.
- Every Environment or Org is uniquely identified by the **Org Id.**
- Sandboxes are largely used to develop, test and freeze the functionality before going live with the customer data thereby reducing the risk factors, minimizing the chances of breaking the existing processes and downtime of the business users and customers.





Different Types of Sandboxes

Any new development cannot be released without testing and development cannot be directly done on Live Env as end users are using the Application.

	 Developer	 Developer Pro	 Partial Data	 Full Copy
Copies Metadata This includes custom settings data	✓	✓	✓	✓
Copies Data Data in your objects such as Accounts & Contacts	✗ ¹ ✓ ²	✗ ¹ ✓ ²	✓	✓
Refresh Interval Minimum amount of time in between refreshes	1 Day	1 Day	5 Days	29 Days
Data Limit Data in your objects such as Accounts & Contacts	200 MB	1 GB	5 GB	Match Prod
Sandbox Templates Pre configured data sampling	✗	✗	✓ ³	✓
Available (Unlimited) Comes with Unlimited Edition of Salesforce⁴	100	5	1	1
Available (Enterprise) Comes with Enterprise Edition of Salesforce⁵	25	0	1	0



Where can I develop? Which Sandbox to choose?

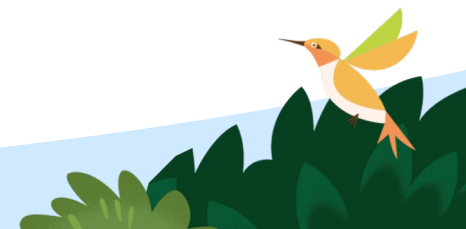
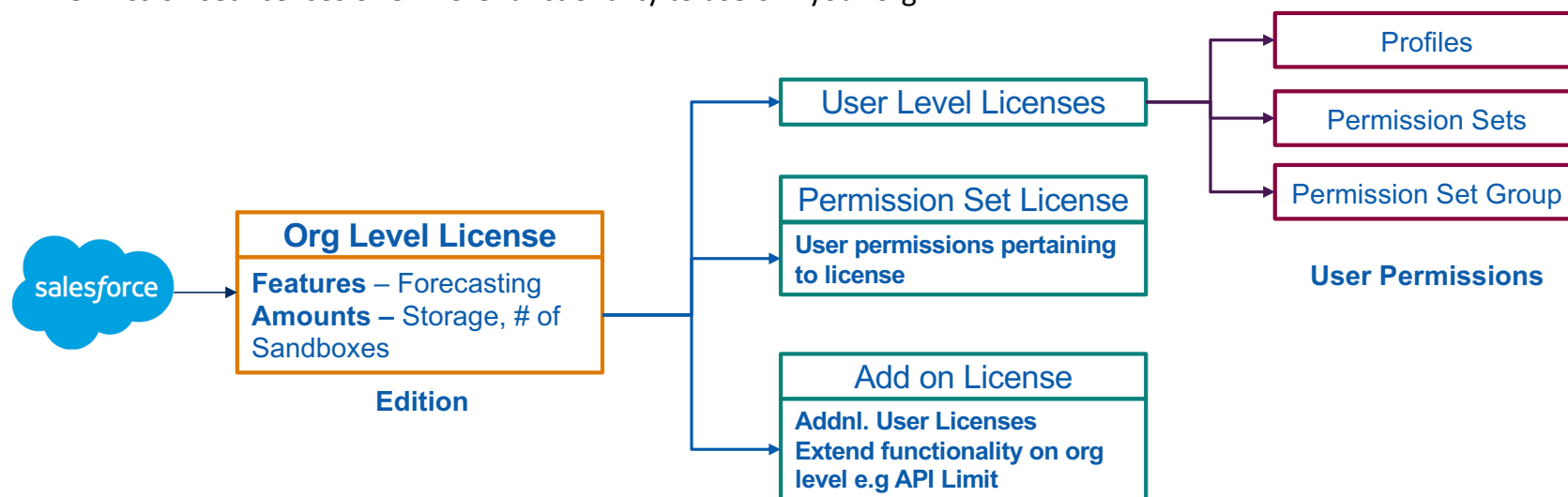
Before making modifications available to end users, we test them in sandboxes to make sure they work as planned and prevent any detrimental effects on production processes.

Sandbox Type	Scratch Org	Developer Sandbox	Developer Pro Org	Partial Copy Org	Full Copy Org
Best for:	Building new functionality	Building new functionality	Feature testing	Feature testing	Integration testing, staging, and UAT
Comments	Meant to last no more than 30 days	Limited data storage	Has more data storage than Developer Sandbox	Contains a subset of production data	Contains the same data as Production



What Happens when you Purchase Salesforce Org

- When you purchase Salesforce, you purchase an org-level license, or edition and you receive licenses, permissions, and profiles that shape your org and what users can do in it.
- To login to Salesforce each user needs a base User License which defines the permissions or the maximum capabilities that a user can have.
- Permission set licenses offer more functionality to users in your org.





Let's review something quickly

How do you order your food @Mc Donald's?

- Based on the frequently ordered items, shops often create and present them as meal.
- Meals are nothing but bundled products which are often preferred by customers.
- Meals not only help customers to be fast in ordering but also appears to be more pocket friendly options.
- Meals can often be bundled based on what you want to choose from. E.g: Any one Burger, Fries and any Beverage of your choice.

SPICY MEALS		MAHARAJA MAC™		ADD TO YOUR MEAL	DESSERTS
BigSpicy™ Wrap		McSpicy™		Piri Piri For JUST ₹21	Phirni McFlurry™ Infused with Rice Bites
Chicken	WRAP MED MEAL ₹182 ₹289 SAVE ₹50	Chicken	BURGER MED MEAL ₹147 ₹254 SAVE ₹50		REGULAR MEDIUM ₹85 ₹99
Paneer	₹177 ₹284 SAVE ₹50	Paneer	₹143 ₹250 SAVE ₹50	CHICKEN McNUGGETS™ 6 PCS 9 PCS ₹123 ₹185	McFLURRY™ Oreo® McFLURRY™ CHOCO CRUNCH SOFT SERVE (HOT FUDGE/STRAWBERRY)
					REGULAR MEDIUM ₹70 ₹97 ₹70 ₹97 ₹74
CLASSIC MEALS		EXTRA VALUE MEALS		BEVERAGES	BREAKFAST
McChicken™		McAloo Tikki™			Egg and Cheese McMuffin™ ₹68
BURGER MED MEAL ₹99 ₹206 SAVE ₹50		BURGER MED MEAL ₹41 ₹148 SAVE ₹50			VEG McMuffin™ ₹81
					DOSA MASALA BRIOCHE (SINGLE PATTY) ₹48
McVeggie™		Chicken Kebab Burger		SOFT DRINKS SMALL MEDIUM LARGE ₹68 ₹74 ₹83 ICED TEA ₹44 ₹70 COLD COFFEE ₹85	
BURGER MED MEAL ₹90 ₹197 SAVE ₹50		BURGER MED MEAL ₹74 ₹181 SAVE ₹50			

Salesforce Editions > Overview



Bundles of features and Services designed for serving specific business needs.

Starter Suite The simple CRM suite with marketing, sales, and service.	Professional The CRM for sales.	Enterprise The CRM for sales with more flexibility and web API.	Unlimited The CRM for sales with full suite of AI, automation, and developer support built-in	Unlimited+ The best of Salesforce for Sales.
\$25 user/month USD (billed monthly or annually)	\$80 user/month USD (billed annually)	\$165 user/month USD (billed annually)	\$330 user/month USD (billed annually)	\$500 user/month USD (billed annually)
✓ Simplified Setup & Onboarding ✓ Lead, Account, Contact & Opportunity Management ✓ Email Integration & Automated Activity Capture	✓ Forecast Management ✓ Customizable Reports & Dashboards ✓ Quoting & Contracting	✓ Everything in Professional plus Advanced Pipeline Management & Deal Insights ✓ Territory Management & Planning ✓ Workflow & Approvals Automation	✓ Everything in Enterprise plus Advanced Automation & Sales Engagement ✓ Predictive & Generative AI ✓ Premier Success Plan & Full Sandbox	✓ Everything in Unlimited plus Performance Management, Enablement and Team Collaboration with Slack ✓ Connect Salesforce and External Data with Data Cloud and Revenue Intelligence
Try for free	Try for free	Try for free	Try for free	Contact us



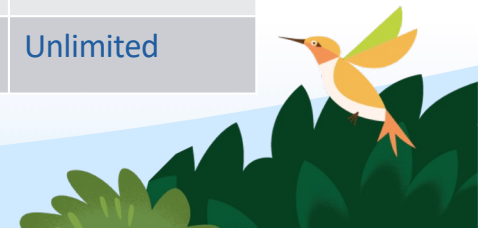
Salesforce Editions > Features

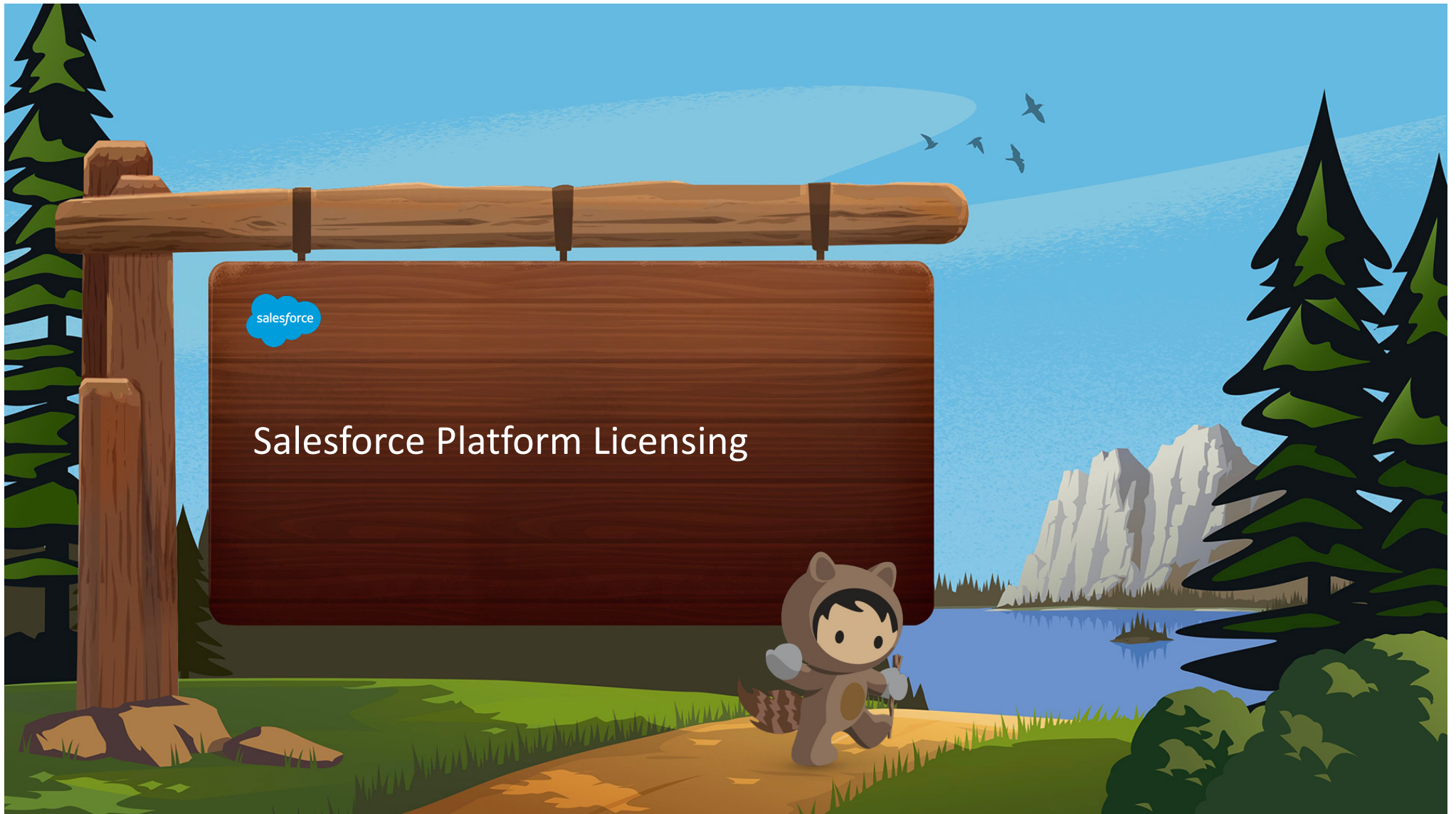
Edition	Features Offered
Essentials	Designed for small businesses getting started with CRM to boost sales or service productivity. It includes a setup assistant and administration tools to customize your deployment as you grow.
Starter	fastest way to get started with the Salesforce CRM suite. It's perfect if you're a first-time user who wants to find, win, and keep customers and build stronger relationships from day one.
Professional	Designed for businesses requiring full-featured CRM functionality. It includes straightforward and easy-to-use customization, integration, and administration tools to facilitate any small to midsize deployment.
Enterprise	Meets the needs of large and complex businesses. gives advanced customization and administration tools, in addition to all the functionality available in Professional Edition, that can support large-scale deployments. Also includes access to Salesforce APIs, so you can easily integrate with back-office systems.
Unlimited	Maximizes your success and extends it across the entire enterprise through the Lightning Platform. Enables platform flexibility for managing and sharing all information on demand. Includes all Enterprise Edition functionality, Premier Support, full mobile access, unlimited custom apps, increased storage limits, and other features
Developer	Provides access to the Lightning Platform and APIs. It lets developers extend Salesforce, integrate with other applications, and develop new tools and applications. Developer Edition also provides access to many of the features available in Enterprise Edition.

Salesforce Features and Edition



	Developer	Essentials	Professional	Enterprise	Unlimited
Active Lookup Filters	5 per object	5 per object	5 per object	5 per object	5 per object
Active Validation Rules	100 per Object	20	20	100	500
Custom Objects	400	0	50	100	2000
Custom Fields Per object	500	100	100	500	800
Max Relationships	40	0	40	40	40
Custom Profiles	1500 per user license type	2	3	1500	1500
Permission Sets	1000	5	10	1000	1000
Permission Set Group	800	0	10	800	800
Sharing Rules per object	300	300	300	300	300
Flows and Processes	2000	5	5	4000	4000
User	2	5	Unlimited	Unlimited	Unlimited







Licensing in Salesforce

Salesforce features and services that are available to your org & a user. ***Licenses are Provisioned.***

An org typically has multiple licenses that together comprise all its available features and services.

An org's licenses include information specific to the org, such as start and end dates, the number of users who can access Salesforce, the number of custom objects the org can create, and so on.

User Licenses

A user license determines the baseline of features that the user can access.

Every user must have exactly one user license.

You assign user permissions for data access through a profile and optionally one or more permission sets.

Permission Set Licenses

Permission set licenses entitle users to access additional features not included in their assigned user license.

Users can be assigned any number of permission set licenses.

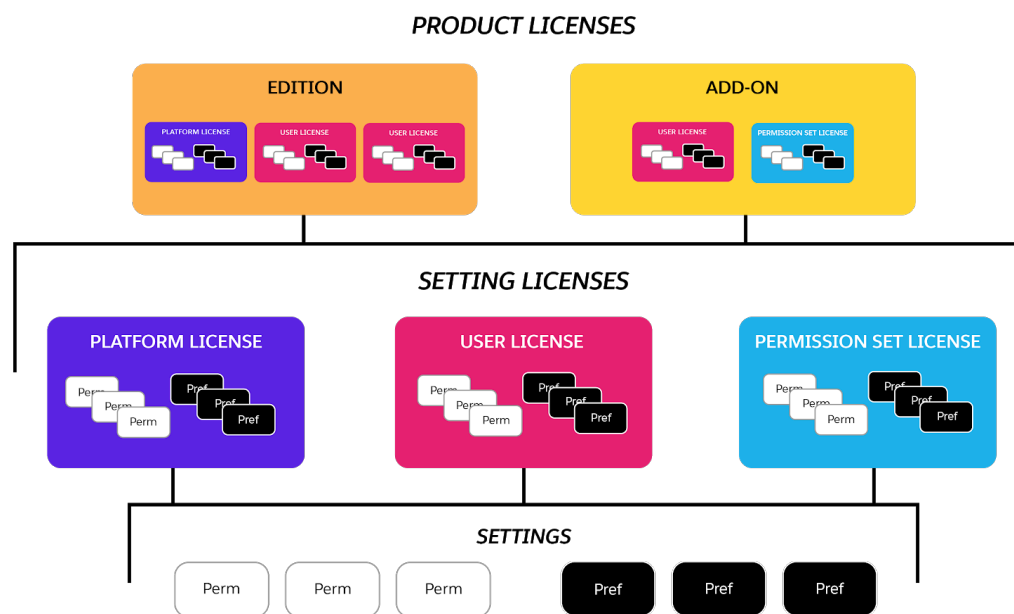
Feature Licenses

A feature license entitles a user to access an additional feature that isn't included with his or her user license, such as Marketing or WDC.

Users can be assigned any number of feature licenses.



Editions and Add-ons



- As a Salesforce customer, you don't directly buy individual perms or prefs.
- Not even the individual platform, user, or permission set licenses (which are all settings licenses).
- Customer buy editions and add-ons, which are product licenses.
- Editions and add-ons bundle together platform, user, and permission set licenses in varying combinations to provide suites of functionality, so that user can easily select the package of features and services needed.
- **While an org must have an edition, it can have any number of AddOns or no AddOns at all.**

Each org has one edition, which provides all the functionality needed to activate the org.

The edition is determined by the org's edition license, which is made up of a grouping of platform and user licenses.

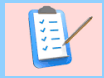


Main License Types

Salesforce offers a variety of license types that provide different levels of access and functionality.

Salesforce User Licenses	Determine access to baseline features.
Salesforce Feature Licenses	Provide additional features not included in the user license.
Salesforce Platform Licenses	For Enterprise and Unlimited editions, include Accounts, Contacts, Reports, and automation tools.
Salesforce Identity Licenses	Enable Salesforce single sign-on access.
Salesforce Einstein Licenses	Integrate AI technology into Salesforce products.
Salesforce Experience Cloud Licenses	Offer external user access to Salesforce org.
Marketing Cloud Licenses	Tailored for various marketing needs.
Pardot Licenses (Account Engagement)	Focus on B2B marketing automation.





Discovery Questions for Developers

Ask these questions to your Team Lead or Project Managers when you are newly onboarded to any Project

- ✓ Project Development and Deployment Lifecycle
- ✓ Sandbox/Environment Strategy
- ✓ Org Edition and Product client is using e.g: Sales Cloud, Industry Cloud
- ✓ Sandbox refresh strategy
- ✓ Release Lifecycle for deployments
- ✓ Tools used for Deployments and Branching strategy
- ✓ Code Review Process



Salesforce Trailhead

Free Learning Platform Provided by Salesforce

Create a new developer org for this training.

Not required to be linked to the Trailhead account.



Creating & Setting Your Trailhead Salesforce Account

This guide provides step-by-step instructions on how to create and set up a Trailhead Salesforce account. It covers everything from verifying your email to completing your profile and uploading a professional picture. By following this guide, you can ensure that your account is properly set up and ready to start earning badges, certifications, and points on Trailhead.



How to Create a Salesforce Developer Account

This guide provides step-by-step instructions on how to create a Salesforce Developer Account. Developer account will never expire.

Sheetal | 10 Steps | 37 seconds



THANK Y☁U

